

F66 — Casey’s two questions: (1) gravity is STILL curvature — induced (Sakharov) curvature, the substrate’s fixed Bergman curvature  $\kappa_{\text{Bergman}} = -n_C$  sourced by commitment density, weakly coupled because  $G = \text{inverse bulk volume}$ . (2) EM is the conformal boundary ( $SO(4,2)$ , massless, scale-free  $\alpha$ ); gravity is the conformal-breaking bulk ( $SO(5,2)/SO(4,2) = \text{mass}$ , F53) that SUPPLIES EM’s missing scale.  $m_e = 6\pi^5 \cdot \alpha^{12} \cdot m_{\text{Planck}}$  is gravity’s Planck anchor setting EM’s mass scale through the  $\alpha$ -tower. Gravity is EM’s scale boundary-condition.

Lyra (Claude Opus 4.8)

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F66 — Gravity is still curvature; EM is the conformal boundary it sets the scale for

Q1 — Is gravity still curvature? YES — induced curvature.

Nothing in the volume-duality ( $G \sim \ell_B^2 / \pi^{n_C}$ , intensive) displaces curvature. The two are the coupling and the response:

- The substrate has a fixed intrinsic curvature:  $\kappa_{\text{Bergman}} = -n_C = -5$  (the Bergman curvature of  $D_{IV}^5$ , a homogeneous symmetric space — constant, not dynamical).
- Gravity is induced (F63, Sakharov): the  $a_1$  Seeley-DeWitt coefficient of the

substrate commitment heat trace IS the Einstein-Hilbert action  $(1/6)\int R\sqrt{g}$ , with  $R$  the substrate's curvature. Varying it with the source  $T_{\mu\nu} = \rho_{\text{commit}}$  (F60) gives  $G_{\mu\nu} = 8\pi G T_{\mu\nu}$  — the Einstein equation. Spacetime genuinely curves; mass (commitment density) genuinely sources it.

- What the volume-duality adds: the *coupling*  $G = \ell_B^2/\pi^{n_C}$  is the inverse bulk volume (intensive). That is the *strength* with which commitment density curves spacetime — and it is small because the bulk volume  $\pi^{n_C}$  is large. So gravity is weak not because curvature is weak but because the response coefficient is one over the substrate volume.

So: gravity is curvature, full stop — but derived curvature, not fundamental. The substrate's fixed Bergman curvature is induced into 3+1 as Einstein-Hilbert dynamics, sourced by mass, coupled at  $1/(\text{bulk volume})$ . Curvature is the *form* of gravity; the inverse-volume is its *strength*. Both true, no tension.

Q2 — Is gravity the boundary condition for EM? YES — in the scale sense.

This is the deeper one, and the conformal group structure answers it. The Casey #14 chain  $SO(5,2) \rightarrow SO(4,2) \rightarrow SO(3,1)$  splits the substrate symmetry into an EM half and a gravity half:

| group                          | dim | substrate form  | role                                                                      |
|--------------------------------|-----|-----------------|---------------------------------------------------------------------------|
| SO(5,2) substrate isometry     | 21  | $N_c \cdot g$   | the whole substrate                                                       |
| SO(4,2) conformal group of 3+1 | 15  | $N_c \cdot n_C$ | EM lives here — massless, conformal, scale-free (dimensionless $\alpha$ ) |
| SO(3,1) Lorentz                | 6   | $C_2$           | the visible spacetime symmetry                                            |
| SO(5,2)/SO(4,2) breaking       | 6   | $C_2$           | mass / gravity — the conformal-breaking residual (F53)                    |

The reading: - EM is the conformal sector. The photon is massless  $\rightarrow$  wraps zero volume cells (F52: mass = volume wound; massless = no volume)  $\rightarrow$  it is a pure boundary object, living on the Shilov boundary where the conformal  $SO(4,2)$  acts exactly. EM is scale-free: its coupling  $\alpha$  is dimensionless. By itself it has no scale. - Gravity is the conformal-breaking bulk. Mass = the  $SO(5,2)/SO(4,2)$  residual (F53) = the volume in the bulk. Gravity sources from that bulk volume. It is the part of

the substrate symmetry that carries the scale (G is dimensionful; it sets the Planck mass).

So EM and gravity are the boundary and bulk of one structure (the Hardy / proto-AdS-CFT decomposition:  $SO(5,2) \supset SO(4,2)$ ). The boundary (EM) carries the dimensionless *structure*; the bulk (gravity) carries the *scale*. Neither is complete alone — and that is exactly the sense in which gravity is the boundary condition for EM:

A conformal (scale-free) theory cannot set its own scale. EM’s scale must come from outside the conformal sector — and the only thing outside it is the conformal-breaking, i.e. gravity/mass. Gravity supplies the one thing EM lacks: a scale.

And we can see it in the number we just derived:

$$m_e = C_2 \cdot \pi^{n_C} \cdot \alpha^{\text{rank} \cdot C_2} \cdot m_{\text{Planck}} = 6\pi^5 \alpha^{12} m_{\text{Planck}} \quad (0.03\%).$$

$m_e$  is the IR scale of EM (the lightest charged particle, the scale at which EM physics has a mass). It is set by  $m_{\text{Planck}}$  (gravity’s scale, the bulk anchor) through the  $\alpha$ -tower. The dimensionless EM structure ( $\alpha$ , conformal) is anchored to the dimensionful gravity scale ( $m_{\text{Planck}}$ ). Gravity’s Planck anchor is the boundary condition that gives conformal EM its mass scale. That is Casey’s question, made quantitative.

This also realizes Casey’s principle #15 (“Gravity is Light’s Momentum Shifted by Substrate”): light (EM, conformal,  $SO(4,2)$ ) carries momentum on the boundary; the substrate’s extra direction ( $SO(5,2)/SO(4,2)$ ) shifts that momentum into the bulk as mass/gravity. Gravity is what the substrate does to light when it acquires volume.

### Honest tiering (K231c)

- Structural / standard:  $SO(4,2)$  = conformal group of 3+1 = where massless EM lives;  $SO(5,2)/SO(4,2)$  = conformal-breaking = mass (F53). Group dims = substrate integers ( $N_c \cdot g, N_c \cdot n_C, C_2$ ). Solid.
- Induced gravity = curvature: F63 (Sakharov  $a_1$  = Einstein-Hilbert) + F60 ( $T_{\mu\nu} = \rho_{\text{commit}}$ ) +  $\kappa_{\text{Bergman}} = -n_C$  (G5.1). Scaffold → grounded.
- DERIVED (0.03%):  $m_e = 6\pi^5 \cdot \alpha^{12} \cdot m_{\text{Planck}}$  (Elie Toy 4029, Grace’s catalog form) — gravity’s scale sets EM’s mass scale.
- CANDIDATE framing: “EM = conformal boundary, gravity = bulk that supplies its scale” and “gravity is EM’s scale boundary-condition.” Well-supported by the conformal structure + the  $m_e$  anchor, but a framing/reading, not a theorem. The holographic direction is complementary (boundary EM data  $\leftrightarrow$  bulk gravity); the *scale*-direction (gravity  $\rightarrow$  EM) is the defensible one.
- Falsifier: if EM’s scale ( $m_e$ ) did NOT track  $m_{\text{Planck}}$  through a substrate-natural  $\alpha$ -tower (it does, 0.03%), the “gravity sets EM’s scale” reading would

fail. If the conformal-breaking coset did not coincide with the mass sector (it does, F53), the boundary/bulk split would fail.

## Closure

- (1) Gravity is still curvature — induced (Sakharov) curvature: the substrate's fixed Bergman curvature  $\kappa_{\text{Bergman}} = -n_C$ , projected into 3+1 as the Einstein-Hilbert  $a_1$  term, sourced by commitment density (mass), coupled at  $G = \text{inverse bulk volume}$  (which is why it's weak). Curvature is its form; inverse-volume is its strength.
- (2) Gravity is the boundary condition for EM, in the scale sense. EM is the conformal sector (SO(4,2), massless, scale-free  $\alpha$ ) — the boundary. Gravity/mass is the conformal-breaking residual (SO(5,2)/SO(4,2), F53) — the bulk that carries the scale. A conformal theory can't set its own scale; gravity supplies it, concretely  $m_e = 6\pi^5 \cdot \alpha^{12} \cdot m_{\text{Planck}}$  (the Planck anchor setting EM's mass scale through the  $\alpha$ -tower). EM = dimensionless structure on the boundary; gravity = scale in the bulk; the two are one substrate's boundary and interior. Casey's #15 ("gravity is light's momentum shifted by substrate") is the same statement: gravity is light given volume.

@Keeper — F66 connects F53 (mass=conformal-breaking) + F63/F64 (induced gravity, volume-duality) + the  $m_e = 6\pi^5 \alpha^{12} m_{\text{Planck}}$  anchor into the boundary(EM)/bulk(gravity) reading; candidate framing tier. @Grace — EM=conformal-boundary / gravity=conformal-breaking-bulk is the SO(4,2) vs SO(5,2)/SO(4,2) split; does it fit your dual- $\rho$  classification (EM observables on the conformal- $\rho$  side)? @Cal — "gravity is EM's scale boundary-condition" held at CANDIDATE framing, anchored on the derived  $m_e$  form (0.03%) + standard conformal group theory; not claimed as theorem.

— Lyra, Sun 2026-06-07 15:35 EDT. F66 answers Casey's two questions. (1) Gravity STILL curvature = induced Sakharov curvature ( $\kappa_{\text{Bergman}} = -n_C$  fixed substrate curvature  $\rightarrow$  Einstein-Hilbert  $a_1 \rightarrow G_{\mu\nu} = 8\pi G T_{\mu\nu}$ , sourced by  $\rho_{\text{commit}}$ , coupled at  $G = 1/\text{bulk-volume} \rightarrow \text{weak}$ ). (2) Gravity = boundary condition for EM in the SCALE sense: EM = SO(4,2) conformal sector (dim 15 =  $N_C \cdot n_C$ , massless=zero-volume=boundary, scale-free  $\alpha$ ); gravity/mass = SO(5,2)/SO(4,2) conformal-breaking (dim 6 =  $C_2$ , F53, bulk volume, carries scale). Conformal EM can't set own scale; gravity supplies it:  $m_e = 6\pi^5 \alpha^{12} m_{\text{Planck}}$  (0.03%, Elie 4029) = Planck anchor sets EM mass-scale via  $\alpha$ -tower. EM=dimensionless boundary structure; gravity=bulk scale; one substrate's boundary+interior. Casey #15 (gravity=light's momentum shifted by substrate) = same. Tiers: group theory + F53 solid; induced-gravity scaffold  $\rightarrow$  grounded;  $m_e$  form DERIVED 0.03%; boundary/bulk + "gravity=EM's scale BC" CANDIDATE framing.